



Proactive aircraft turnaround buffer optimization integrating machine learning and scenario analysis

Konstantia Kontodimou^{a,*}, Dionysios Peteinatos^a, Stavroula Bagioneta^b,
Konstantinos Kepatsoglou^a

^a Laboratory of Transportation Engineering, National Technical University of Athens, Athens, 157 80

^b Department of Civil and Environmental Engineering, West Virginia University, Morgantown, WV 26505

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ABSTRACT

Turnaround uncertainty is a major contributor to airline schedule unreliability and the propagation of delays across flight networks. This study proposes a learning-enhanced optimization framework for proactive aircraft turnaround buffer allocation that integrates data-driven delay predictions into planning decisions. Flight-level turnaround risk is estimated using an Extreme Gradient Boosting (XGBoost) regression model trained on historical operational data, with uncertainty captured through scenario generation based on prediction residuals. These scenarios are incorporated into a stochastic mixed-integer linear programming model that determines how limited buffer time should be allocated across flights to balance expected delay costs, buffer-related utilization costs, and environmental impacts associated with extended ground operations. The proposed framework is evaluated using real operational data from multiple airlines and aircraft types. The results show that optimal turnaround buffering is highly heterogeneous and strongly influenced by aircraft utilization patterns and exposure to delay accumulation, rather than uniform scheduling rules or airline size. Sensitivity analysis further demonstrates that buffer allocation responds coherently to economic trade-offs between delay penalties and schedule efficiency, with nonlinear transitions across buffering regimes. Overall, the study illustrates how machine-learning-derived uncertainty can be embedded into prescriptive optimization models to support proactive turnaround planning. The proposed approach provides a practical predictive-prescriptive decision-support tool for improving robustness in airline ground operations under uncertainty.

1. Introduction and background

Turnaround performance is a central driver of airline schedule reliability and a major contributor to delay propagation across flight networks. The aircraft turnaround, comprising passenger boarding and deboarding, cabin cleaning, refueling, catering, baggage handling, and apron coordination, represents a tightly coupled sequence of ground activities whose variability directly affects departure punctuality and downstream operational stability. Beyond its operational significance, turnaround represents a critical segment of ground handling that industry stakeholders increasingly recognize as integral to both safety and efficiency: recent regulatory efforts by the European Union Aviation Safety Agency emphasize that ground handling activities have direct implications for flight safety and must be standardized and overseen to ensure safe and reliable operations (EASA, 2025), while the

International Air Transport Association highlights ground operations, including turnaround coordination, procedural standardization, and ground support equipment management, as a priority area for improving on-time performance and operational resilience (IATA, 2025).

Foundational reviews emphasize that turnaround variability is primarily driven by passenger-related processes and the coordination of parallel ground handling tasks, rather than by purely technical servicing times (Schmidt, 2017). From a design and planning perspective, simulation-based studies show that boarding and deboarding alone may account for up to 40 % of total turnaround duration, showing that turnaround considerations are relevant even at early stages of aircraft design (Picchi Scardaoni, Magnacca, Massai & Cipolla, 2021). These findings align with process-oriented and managerial case studies that frame turnaround time as a strategic lever for airline competitiveness,

* Corresponding author at: Laboratory of Transportation Engineering, National Technical University of Athens, 9, Iroon Polytechniou Str, 157 80 Athens, Greece.

E-mail address: k_kontodimou@mail.ntua.gr (K. Kontodimou).

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particularly for high-utilization and low-cost operating models, where even marginal reductions in ground time translate into significant cost savings and fleet productivity gains (More & Sharma, 2014).

A complementary stream of research focuses on coordination and information sharing during turnaround operations. Empirical evidence from Airport Collaborative Decision Making (A-CDM) environments shows that structured information exchange among airlines, airports, air navigation service providers, and ground handlers improves turnaround predictability, resource utilization, and schedule adherence (Okwir et al., 2017). At the airport-system level, workforce availability and coordination have been identified as critical enablers of turnaround performance, with recent literature documenting how staffing shortages, retention challenges, and limited digitalization can amplify operational disruptions (Wandelt & Wang, 2024). These insights highlight that turnaround uncertainty is not only technical but also organizational and systemic in nature.

Recent technological contributions explore automation, digital twins, and smart apron concepts, suggesting potential efficiency gains through real-time monitoring and synchronization of ground processes (Lu et al., 2025). Related analytical work proposes deterministic and algebraic frameworks to identify critical temporal thresholds during ground handling, showing that small disturbances at specific “critical minutes” can trigger nonlinear delay propagation (Tuduriu et al., 2025). While these approaches provide valuable structural and diagnostic insights, they remain largely simulation-driven or analytical and offer limited guidance on how predicted variability should be translated into proactive planning decisions.

Complementary literature addresses robustness and buffering in airline operations. Empirical and analytical studies demonstrate that ground buffers are effective in mitigating propagated delays and can substitute for upstream flight buffers depending on cost–variability trade-offs (Brueckner, Czerny & Gaggero, 2021). Robust optimization models in aircraft routing and maintenance planning similarly exploit turnaround-related slack or time reductions to improve delay absorption and fleet productivity under uncertainty (Eltoukhy et al., 2020). Broader reviews of robust airline scheduling emphasize the importance of explicitly accounting for turnaround uncertainty when designing resilient schedules, particularly in environments where multiple stakeholders jointly influence operational outcomes (Guardo-Martinez et al., 2026).

At the system level, integrated scheduling and operational control frameworks demonstrate that delay reductions can be achieved by jointly coordinating schedule adjustments, aircraft assignment, and operational resource utilization, rather than treating these decisions sequentially (Jacquillat & Odoni, 2015; Glomb, Liers & Rösel, 2023). State-of-the-art reviews of airline integrated scheduling further emphasize that robustness and operational consistency depend on capturing interdependencies across planning stages and resources, including aircraft, crews, and ground operations (Xu, 2024). However, in most integrated models, buffer decisions remain static, uniform, or weakly differentiated across flights.

Predictive and prescriptive analytics, including machine learning and optimization, have been increasingly applied across airline operations and air transportation systems. Machine learning models have demonstrated strong performance in forecasting turnaround milestone times within Airport Collaborative Decision-Making A-CDM settings, enabling more accurate anticipation of operational delays (Tang et al., 2025). Parallel developments in transportation science integrate predictive and prescriptive analytics for air traffic flow management and airline disruption management, including passenger-centric ground delay programs (Jacquillat, 2021) and dynamic recovery models under airport operating uncertainty (Lee, Marla & Jacquillat, 2020). In a related predictive–prescriptive contribution, machine-learning-based primary delay predictions are embedded within an aircraft routing optimization framework, demonstrating how probabilistic delay forecasts can be integrated into network-level aircraft assignment and

rotation decisions to mitigate aircraft delays for a specific airline (Biolini & Jacquillat, 2023). In addition, branch-and-cut approaches have been proposed for airline schedule generation problems (Erdmann, Nolte, Noltemeier & Schrader, 2001). In an industrial Air France setting, compact integer programming formulations have been developed for aircraft routing, alongside cut-generating frameworks for integrated aircraft-routing and crew-pairing problems (Parmentier & Meunier, 2020). Likewise, the use of integer programming formulations solved with off-the-shelf commercial solvers has been highlighted as particularly valuable in airline planning, as it enables prototyping, testing, and integration of new planning concepts into broader decision-support frameworks (AhmadBeygi, Cohn & Lapp, 2010). Conceptual frameworks further explore how artificial intelligence can support collaborative decision-making in airline disruption management by enhancing data-driven coordination and system-level performance monitoring (Geske, Herold & Kummer, 2024). At a broader level, reviews of artificial intelligence in air transportation and air traffic management document the rapid transition from rule-based systems to learning-based prediction and optimization across ATM, airline, and airport operations (Du, Wu, Leng & Wandelt, 2025). These studies highlight the growing potential of AI-enabled decision support while noting persistent challenges related to integration, interpretability, and operational deployment.

Taken together, the literature reveals three dominant but largely separate research directions. First, turnaround processes and buffers are extensively analyzed using descriptive, empirical, simulation-based, or analytical approaches, often focusing on diagnosis rather than prescriptive allocation. Second, machine learning models are increasingly used to predict turnaround durations and delay risks, yet these predictions are rarely embedded directly into optimization-based planning decisions. Third, optimization models for airline scheduling, robustness, and disruption management incorporate uncertainty at various levels, but turnaround time variability is typically treated in an aggregated or static manner, rather than as a flight-specific, data-driven source of risk.

Existing predictive–prescriptive studies demonstrate the value of probabilistic forecasts within optimization frameworks, particularly in air traffic flow management and disruption recovery contexts where decisions are made reactively or at the network level. Most existing contributions focus on network-level decision variables such as aircraft routing, departure timing adjustments, slot allocation, or disruption recovery mechanisms, where delay propagation is managed through schedule redesign. Proactive planning decisions related to the allocation of limited turnaround buffer time across individual flights, explicitly informed by predicted turnaround risk and subject to system-wide constraints, have received comparatively limited attention. In particular, the literature has not extensively examined how limited turnaround slack should be distributed across flights within a fixed daily schedule under a total buffer allocation constraint.

This paper contributes to this research area by proposing a learning-enhanced stochastic optimization framework for aircraft turnaround buffer allocation. Using machine learning predictions of required turnaround duration and associated uncertainty, the model determines how limited buffer minutes should be optimally distributed across flights to balance expected delay costs, buffer-related utilization costs, and a system-level constraint on total available buffer time. In contrast to routing-based or schedule-adjustment approaches, the present framework assumes fixed departure times, aircraft assignments, and network structure, and treats the allocation of turnaround slack as the sole endogenous decision variable. Unlike fixed or heuristic buffering strategies, the proposed approach is flight-specific, risk-aware, and explicitly coupled across the schedule through a buffer budget, providing a proactive decision-support tool for turnaround planning aligned with recent advances in data-driven airline operations. Furthermore, the formulation explicitly models timing relationships across rotation-connected flights and incorporates an environmental proxy associated with ground occupancy due to Auxiliary Power Unit (APU) usage,

thereby extending predictive-prescriptive modelling beyond delay minimization alone.

By isolating the slack allocation problem from broader schedule redesign or disruption recovery mechanisms, the proposed framework addresses a distinct decision layer within airline operations, namely, the pre-operational distribution of limited turnaround resources under uncertainty. The explicit coupling of machine-learning-derived turnaround uncertainty with a buffer budget constraint, cost trade-offs, and an environmental component within a flight-level buffer allocation problem represents a novel contribution distinct from prior predictive-prescriptive scheduling frameworks.

In this study, the system boundary is deliberately defined at the fleet-day level, and the problem is formulated as a tactical buffer-allocation decision within an already feasible operating plan. Specifically, we assume that the published schedule and associated fleet assignment have already satisfied the core feasibility requirements related to aircraft routing, maintenance planning, passenger itineraries, and crew duty regulations. Under this assumption, these elements are treated as upstream constraints rather than endogenous decision variables. Accordingly, the objective of the proposed framework is to evaluate how robustness can be improved within a fixed timetable through the allocation of additional turnaround buffer time under delay uncertainty. This reflects a common operational setting in airline operations control, where short-horizon adjustments are implemented without reopening the full schedule design problem.

The remainder of this paper is organized as follows. The next section presents the proposed learning-enhanced modeling framework, describing the predictive component used to estimate flight-level turnaround duration and uncertainty, and the formulation of a stochastic mixed-integer linear programming (MILP) model for turnaround buffer allocation. It details the key parameters, including baseline turnaround times, scenario-based delay realizations, cost coefficients, and the system-wide buffer budget that couples decisions across flights. The subsequent section reports the computational results based on real operational data, illustrating how optimal buffer allocations vary across airlines and aircraft types, and examining the sensitivity of buffering decisions to delay and buffer cost assumptions. Finally, the paper concludes by discussing the implications of the proposed framework for proactive turnaround planning and by outlining directions for future research, including dynamic re-optimization and deeper integration with other airline planning and operational decisions.

2. Methodology

This study adopts a predictive-prescriptive framework in which machine-learning-based delay predictions are embedded into a mixed-integer linear optimization model for proactive turnaround buffer allocation. The methodology consists of two main components: (i) prediction of downstream delays using gradient-boosted trees, and (ii) optimization of buffer allocation using a stochastic MILP formulation that trades off planned reserve time against expected delay costs.

The predictive component estimates the delay risk using historical operational data, a supervised learning dataset is constructed where the target variable is the realized downstream delay of a flight, capturing delay spillovers. An Extreme Gradient Boosting (XGBoost) regression model is employed due to its strong predictive performance for nonlinear relationships and its robustness in handling heterogeneous operational data.

The target variable in the predictive model is the realized arrival delay per flight, obtained from the Bureau of Transportation Statistics (BTS) On-Time Performance dataset ([U.S. Bureau of Transportation Statistics \(BTS\) 2025](#)). This variable represents the total operational delay experienced by each flight and inherently aggregates multiple sources of disruption, including carrier delays, weather-related delays, NAS (air traffic management) delays, security delays, and late-aircraft delays. Therefore, the prediction task addressed in this study concerns

total arrival delay rather than isolated primary delays. This distinction is important, as total delay reflects both exogenous uncertainty and endogenous delay propagation effects within the network, making it a more comprehensive but also more complex prediction problem.

The XGBoost model serves as the predictive foundation of the framework, translating operational features into flight-specific estimates of propagated delay risk. Its role is the reliable quantification of heterogeneous turnaround uncertainty that can be embedded into prescriptive optimization decisions. Since the Extreme Gradient Boosting (XGBoost) regression model captures nonlinear relationships between operational flight features and expected downstream delay, it is used to calculate the propagated delay prediction component. Formally, for each flight observation f , the model receives an input feature vector $X_f \in \mathbb{R}^p$, consisting of the following engineered explanatory variables: arrival delay, departure delay, slack time between rotations, flight distance, time-of-week indicators, airline operator, and aircraft type. The target variable is the propagated delay y_f , defined as the nonnegative delay transmitted to the subsequent flight in the aircraft sequence. XGBoost models this relationship through an additive ensemble of regression trees. Specifically, the predicted propagated delay is expressed as:

$$\hat{y}_f = \sum_{k=1}^K g_k(X_f),$$

where each g_k represents a decision tree and K is the total number of boosting iterations. Unlike linear regression, XGBoost does not assign a single global coefficient to each variable; instead, each feature contributes through a sequence of hierarchical split rules across trees. At each boosting step, a new tree is trained to approximate the residual error of the current ensemble, thereby iteratively improving predictive accuracy. The objective minimized during training combines a squared-error loss function with regularization to control model complexity:

$$\mathcal{L} = \sum_f (y_f - \hat{y}_f)^2 + \sum_{k=1}^K \Omega(g_k),$$

where $\Omega(g_k)$ penalizes overly complex trees via constraints on depth, leaf weights, and split structure. Key hyperparameters govern the learning behavior: the learning rate η scales the contribution of each new tree, maximum depth controls the complexity of feature interactions, and the minimum child weight parameter enforces robustness by requiring sufficient data support for splits. Additional stochastic regularization is introduced through subsampling (subsample) and column sampling (colsample_bytree), which reduce overfitting by training each tree on random subsets of observations and features. Furthermore, L2 regularization (λ) and split penalties (γ) discourage unnecessary tree growth, ensuring the model captures meaningful delay patterns rather than noise. Overall, the XGBoost regression framework provides a flexible, nonlinear predictor of propagated delays by sequentially combining many soft decision rules into a strong ensemble model that captures complex operational dependencies in airline turnaround dynamics.

Model inputs include numerical variables capturing the flight date (day of week), realized arrival and departure delays of the inbound flight, flight distance, scheduled slack between arrival and next departure, and arrival and departure clock times converted to minutes. Also, categorical indicators for the operating airline and aircraft type are incorporated and encoded using one-hot encoding.

In addition to schedule and operational variables, the feature set explicitly incorporates weather-related information, including wind speed, visibility, temperature, and humidity. These variables are included to capture exogenous sources of uncertainty affecting flight operations. Combined with the delay target variable, which already reflects realized weather impacts, this allows the model to account for both direct and indirect effects of weather on flight performance.

Feature engineering focuses on preparing operational data so that it can be reliably used for predicting turnaround-related delay risk. Only information available prior to buffer allocation is considered, ensuring that the model relies exclusively on inputs that would be known at the time of planning. This temporal consistency guarantees that the predictive component reflects realistic decision conditions rather than retrospective information. Additional preprocessing steps included conservative data cleaning procedures to remove cancelled flights and incomplete records, enforcing aircraft rotation continuity by aligning inbound and downstream legs chronologically by tail number, and excluding observations that violate rotation consistency.

The dataset is sorted temporally to preserve aircraft sequence integrity. A strict chronological 80/20 train-validation split is applied to preserve temporal integrity and avoid leakage across rotations. Hyperparameters were selected using early stopping based on validation RMSE to control model complexity and reduce overfitting risk.

Uncertainty is represented through an empirical residual-based scenario generation procedure. Residuals are computed exclusively from training-set predictions to prevent information leakage into the stochastic modelling stage. Specifically, residuals are defined as the difference between the observed propagated delay y_f and the corresponding predicted value $\hat{y}_f$ for observations in the training set. Validation residuals are not used in scenario generation, since they originate from a held-out future evaluation period and could introduce subtle look-ahead bias in time-ordered operational settings. Although validation residuals may provide a more conservative representation of out-of-sample uncertainty, their inclusion would compromise the strict separation between model training and downstream optimization. To avoid distortion from rare structural disruptions (e.g., extreme weather or network breakdowns), residuals exceeding ± 180 min are excluded using a symmetric filtering threshold. For each flight in the optimization stage, stochastic realizations are then generated by sampling residuals with replacement from this filtered training residual pool and adding them to the point prediction. This procedure yields a discrete empirical approximation of the predictive distribution that is then embedded into the scenario-based MILP.

It is important to note that the buffer adjustments considered in this study are incremental and operate within the slack already present in the schedule. As such, they do not materially alter aircraft routing structures or long-horizon utilization patterns and therefore are not expected to interfere with maintenance planning requirements. Instead, maintenance feasibility is preserved by design using an already validated fleet-day operating plan.

The modeling approach adopted in this paper is intentionally parsimonious. Since the flight schedule and fleet assignment are fixed, the decision space is restricted to the allocation of buffer time subject to a finite budget. This allows the model to isolate the marginal operational value of buffer redistribution without introducing additional layers of complexity associated with integrated planning.

While integrated scheduling frameworks and decomposition-based stochastic programming approaches are highly relevant for large-scale airline planning problems, they are most appropriate when multiple resource dimensions (e.g., fleet, crew, passengers) must be optimized simultaneously. In contrast, the present problem is narrower in scope and focuses on a single class of tactical decisions. For such problems, compact stochastic mixed-integer linear programming (MILP) formulations offer important advantages in terms of interpretability, transparency, and ease of implementation. For the same reason, we employ a direct solution approach based on branch-and-cut using a commercial MILP solver. This choice is consistent with established practice in airline optimization, where compact formulations of tactical or day-of-operation problems are frequently solved to optimality within operationally acceptable time limits. The size and structure of the proposed model do not necessitate more elaborate decomposition schemes, and the use of an exact off-the-shelf solver facilitates reproducibility and

practical deployment.

The optimization model determines how much turnaround buffer time should be allocated to each flight to balance buffer costs against expected delay impacts. The decision variables, parameters, and sets used in the model are defined as follows:

Sets and indices

F: set of flights, indexed by f

Ω : set of scenarios, indexed by ω

Decision variable

$b_f \geq 0$: planned buffer time (min) added on top of baseline turnaround for flight f

$\delta_{f,\omega}$: Departure delay of flight f under scenario ω (minutes)

$g_{f,\omega}$: Total ground occupancy time of flight f under scenario ω (minutes)

Parameters

T_f^{base} : Baseline technical turnaround time for flight f (minutes)

$D_{f,\omega}$: Required turnaround duration for flight f in scenario ω (obtained from the ML model in minutes)

π_ω : Probability of scenario ω , with $\sum_{\omega \in \Omega} \pi_\omega = 1$

c_f^{delay} : Cost per minute of delay for flight f (€/min)

c^{buf} : Cost per minute of planned buffer (€/min)

c^{env} : Cost per minute of total ground occupancy time (environmental proxy) (€/min)

B^{tot} : Total buffer budget available across all examined flights (minutes)

The objective function is given by:

$$\min Z = \underbrace{\sum_{f \in F} c^{\text{buf}} b_f}_{\text{planned buffer cost}} + \underbrace{\sum_{\omega \in \Omega} \pi_\omega \sum_{f \in F} (c_f^{\text{delay}} \delta_{f,\omega} + c^{\text{env}} g_{f,\omega})}_{\text{expected delay \& environmental cost}} \quad (1)$$

The first term captures the cost of allocating planned buffer time, reflecting reduced aircraft utilization and schedule flexibility. The second term represents the expected delay cost across scenarios, weighted by scenario probabilities and flight-specific delay penalties. The third term introduces an environmental proxy proportional to the total ground occupancy time $g_{f,\omega}$, accounting for emissions associated with prolonged ground operations, primarily from the use of the auxiliary power unit (APU).

The model is subject to the following constraints:

$$\delta_{f,\omega} \geq D_{f,\omega} - (T_f^{\text{base}} + b_f) \quad \forall f \in F, \omega \in \Omega \quad (2)$$

$$g_{f,\omega} = T_f^{\text{base}} + b_f + \delta_{f,\omega} \quad \forall f \in F, \omega \in \Omega \quad (3)$$

$$\sum_{f \in F} b_f \leq B^{\text{tot}} \quad (4)$$

$$b_f \geq 0, \delta_{f,\omega} \geq 0, g_{f,\omega} \geq 0 \quad (5)$$

Constraint (2) ensures that any shortfall between the required turnaround time and the planned ground time materializes as departure delay, while sufficient buffer fully absorbs turnaround uncertainty without delay. In detail, the machine learning model predicts the arrival delay of the next flight operated by the same tail number, denoted as y_f . This value represents the propagated delay transmitted along the aircraft rotation and captures the incremental schedule pressure affecting the subsequent turnaround. The stochastic turnaround requirement used in the optimization model is constructed by combining

the technical baseline turnaround time with the predicted downstream delay and a stochastic residual term. Specifically, it is defined as:

$$D_{f,\omega} = T_f^{\text{base}} + \hat{y}_f + \epsilon_\omega$$

where T_f^{base} denotes the required turnaround duration for flight f , $\hat{y}_f$ is the predicted propagated arrival delay obtained from the XGBoost model, and ϵ_ω represents a sampled residual capturing predictive uncertainty under scenario ω . Thus, the machine learning model does not directly predict the required turnaround duration. Instead, it estimates the expected propagated delay, which is then added to the technical baseline and combined with a sampled residual term to generate the stochastic required turnaround duration $D_{f,\omega}$. This mapping ensures a clear and consistent linkage between the predictive component and the scenario-based optimization model. In addition, constraint (3) defines total ground occupancy time as the sum of the baseline technical turnaround time, the allocated buffer, and any realized delay, linking buffer and delay decisions to overall ground occupancy. Constraint (4) imposes a system-wide buffer budget that couples buffer allocation decisions across flights and induces trade-offs in distributing limited slack among operations with different risk profiles. The variables $\delta_{f,\omega}$ and $g_{f,\omega}$ are endogenous variables determined within the model through the system of constraints. Specifically, $\delta_{f,\omega}$ captures the realized delay that occurs when the allocated buffer (b_f) is insufficient to absorb the stochastic turnaround requirement under scenario Ω . The variable $g_{f,\omega}$ represents the resulting ground time and is defined as a function of the baseline turnaround time, the allocated buffer, and the realized delay. Constraint (5) enforces non-negativity of all decision and recourse variables.

The resulting formulation is a stochastic Mixed-Integer Linear Programming (MILP) model. Although the current implementation involves continuous decision variables, the MILP framework is adopted to ensure extensibility to discrete operational decisions and to enable the computation of globally optimal solutions under the linear problem structure. The optimization problem is implemented in Python using the PuLP modelling interface and solved with the CBC (COIN—OR Branch and Cut) solver. All experiments were conducted on a standard computing machine equipped with 8 GB of RAM and an 11th Generation Intel Core i5–1135G7 processor (4 cores, 8 threads), demonstrating that the proposed approach can be efficiently executed on commonly available hardware without requiring high-performance computing resources.

3. Results

3.1. Data collection

The numerical inputs used in the optimization model were constructed from a combination of publicly available operational data sources and established aviation cost and emissions benchmarks. Flight schedules, arrival and departure times, realized delays, and aircraft tail numbers were obtained from the U.S. Bureau of Transportation Statistics (BTS) TranStats database, which provides detailed, flight-level operational records for scheduled airline operations (U.S. Bureau of Transportation Statistics (BTS) 2025). Aircraft tail numbers were subsequently mapped to specific aircraft models using the FAA aircraft registry to ensure accurate assignment of aircraft characteristics and baseline technical turnaround times (FAA, 2025).

Cost parameters related to operational disruptions were calibrated using standardized aviation cost–benefit analysis inputs. The cost of departure delay per minute and the opportunity cost of planned buffer time were derived from EUROCONTROL’s reference values for delay costs, which account for airline operating impacts, passenger time valuation, and knock-on network effects (EUROCONTROL, 2025). Buffer costs reflect the opportunity cost of increased ground time in terms of reduced aircraft utilization and schedule tightness.

Environmental inputs were constructed by linking total ground occupancy time to auxiliary power unit (APU) fuel consumption and associated carbon emissions. Representative APU fuel burn rates (kg of fuel per hour) were obtained from industry analyses and manufacturer documentation, including operational guidance and aircraft technical characteristics for narrowbody aircraft (Open Airlines, 2024). Fuel burn was converted to CO₂ emissions using standard jet fuel emission factors reported by EUROCONTROL (EUROCONTROL, 2025). The resulting emissions were monetized using estimates of the social cost of carbon based on recent assessments published by the U.S. Environmental Protection Agency, reflecting representative values of climate damage costs per metric ton of CO₂ (EPA) (U.S. Environmental Protection Agency, 2023).

All cost and environmental parameters were expressed on a per-minute basis to ensure consistency with the temporal resolution of the turnaround buffer decision variables. Where parameter values vary across sources or aircraft types, representative mid-range values were selected and subsequently tested through sensitivity analyses to assess the robustness of buffer allocation outcomes to alternative cost assumptions.

3.2. Predictive model

The predictive model is implemented using XGBoost. The final model consists of 1750 trees with a total of 6910 leaves/ parameters. Training was performed using the histogram-based tree method (“hist”), with maximum tree depth of 4, learning rate 0.02, L2 regularization parameter $\lambda = 10$, minimum child weight 25, subsampling rate 0.85, and column sampling rate 0.85. Early stopping with 100 rounds was applied, resulting in 1649 boosting iterations.

The predictive performance of the model should be interpreted considering the scope and data characteristics of this study. While indicative prior work (Biolini & Jacquillat, 2023) report lower prediction errors focusing on delays of one specific airline, the setting differs substantially in terms of objective, data richness, and problem structure. Specifically, the focus is on predicting primary delays, carefully isolated from propagation effects, using proprietary airline data that include detailed operational variables such as aircraft rotations, passenger flows, and congestion measures derived from queuing models (Biolini & Jacquillat, 2023). In contrast, the model presented predicts total arrival delay using publicly available BTS data across multiple airlines and airports over a limited time horizon of one month. This broader and more heterogeneous setting introduces additional variability and restricts access to high-resolution operational features, making the prediction task inherently more challenging.

Using a strict chronological 80/20 train–validation split to preserve temporal integrity, the selected XGBoost model achieves a validation MAE of 9.53 min and a validation RMSE of 32.43 min. On the training set, the model attains a MAE of 9.58 min and an RMSE of 32.36 min. The small gap between training and validation errors indicates good generalization performance. Although the prediction error remains higher than in more controlled or data-rich settings, it is consistent with the increased complexity of predicting total delay in a heterogeneous, real-world environment.

For comparison, the historical mean baseline yields a MAE of 21.55 min and an RMSE of 44.82 min, while a linear regression (Ridge) benchmark achieves a MAE of 19.51 min and an RMSE of 42.51 min. The XGBoost model improves RMSE by approximately 27.6 % relative to the historical mean baseline (from 44.82 to 32.43 min) and by approximately 23.7 % relative to the linear regression benchmark (from 42.51 to 32.43 min). These comparisons demonstrate that nonlinear interactions among inbound delay, slack, aircraft characteristics, and network context improve predictive accuracy beyond simpler specifications.

Beyond prediction accuracy, we explicitly quantify uncertainty through model calibration assessment. Calibration metrics computed on

the validation set show very low Expected Calibration Error (ECE = 0.0032), Maximum Calibration Error (MCE = 0.0147), and Root Mean Squared Calibration Error (RMSCE = 0.0044), indicating that the predictive distribution is well aligned with empirical accuracy. For comparison, a Gaussian Process Regression (GPR) benchmark yields higher error (MAE = 20.15 min, RMSE = 49.28 min) and inferior calibration (ECE = 0.0111; RMSCE = 0.0142), confirming that the proposed XGBoost-based framework provides both superior predictive accuracy and better-calibrated uncertainty estimates.

3.3. Optimization

The stochastic optimization model is solved separately for each airline, aircraft-type, and day instance, resulting in 217 instances in total. A single optimization instance is defined at the level of a daily airline and aircraft-type subproblem. The planning horizon corresponds to a single operational day, and each optimization instance is defined as all rotation-connected flights of a specific airline and aircraft type within that day. The dataset covers all flights, from multiple airports, operated during 31 consecutive calendar days (January 2025) by selected fleets of four U.S. carriers: Delta Air Lines (DL), United Airlines (UA), Frontier Airlines (F9), and Allegiant Air (G4). Aircraft types included are A320 (DL, UA, F9, G4), B737 (DL, UA), and B767 (DL), identified through FAA-registered tail numbers and rotation continuity (arrival airport of flight f equals departure airport of the subsequent flight of the same tail). Depending on the fleet, daily instances include: 178 flights (DL A320), 306 flights (DL B737), 486 flights (DL B767), 157 flights (UA A320), 1181 flights (UA B737), 274 flights (F9 A320), and 185 flights (G4 A320). Within each calendar day, flights are grouped by operator and aircraft type, and the stochastic buffer allocation model is solved for each (airline, aircraft type, day) combination. Thus, an instance consists of all rotation-connected flights operated by a specific airline with a specific aircraft family on a given day. Flights are included only if they belong to the predefined tail-number set, are not cancelled, have valid timing information, and form feasible aircraft rotations (i.e., the destination of one flight matches the origin of the subsequent flight for the same tail number).

In addition, the stochastic optimization model is implemented using a discrete scenario set Ω constructed via residual bootstrapping. In the experiments, the number of scenarios is fixed to $|\Omega| = 20$. Each scenario is assigned an equal probability weight, with

$$\pi_\omega = \frac{1}{|\Omega|}, \forall \omega \in \Omega$$

corresponding to a uniform empirical approximation of the predictive error distribution. These probabilities are explicitly incorporated into the objective function through the expected delay and environmental cost terms. To assess robustness with respect to the number of scenarios, we conducted a sensitivity analysis over $|\Omega| \in \{5, 10, 20, 40, 60, 80, 100\}$. The resulting average optimized mean buffer and corresponding solve times are summarized in Table 1.

The results indicate that buffer allocations stabilize rapidly and results are not sensitive to $|\Omega|$. Increasing $|\Omega|$ beyond 20 produces negligible variation in the average optimized buffer (less than 0.5 min

difference), while computational time increases approximately linearly. Based on this convergence behavior, $|\Omega| = 20$ was selected as a computationally efficient and statistically stable baseline.

Regarding the number of scenarios, the limited variation observed between solutions obtained with $|\Omega| = 5$ and $|\Omega| = 20$ indicates that the stochastic program is stable and that the key characteristics of uncertainty are already captured with a moderate number of scenarios. This behavior is consistent with sample average approximation approaches, where solution convergence with respect to scenario size is a desirable property. Therefore, the marginal differences observed when increasing the number of scenarios should not be interpreted as a lack of stochastic variability, but rather as evidence that the model captures the dominant uncertainty patterns efficiently without requiring a large scenario set.

Across instances, the number of flights per instance ranges from 10 to 1298, with a mean of 396 flights and a 95th percentile of 1244 flights. Under the baseline configuration with $|\Omega| = 20$ scenarios, the resulting deterministic equivalent formulation contains between 410 and 53,218 decision variables (mean: 16,219; 95th percentile: 50,988) and between 401 and 51,921 linear constraints (mean: 15,824; 95th percentile: 49,745).

Beyond model dimensionality, a key design parameter of the framework is the total buffer budget B_{tot} , which represents the aggregate amount of turnaround slack that can be allocated across flights within each daily optimization instance. Since airline-specific buffer capacity or managerial slack allocation policies are not publicly available, B_{tot} is treated as a calibration parameter rather than a fixed externally observed operational constant. To ensure real-world relevance and robustness, we conducted a sensitivity analysis over a wide range of budget values (300 to 30,000 min), as shown in Fig. 1. The results indicate that optimized mean buffer allocations increase gradually for small budgets but reach a saturation level beyond approximately $B_{tot} = 6000$ min, after which additional budget produces negligible changes in optimal buffer allocations. The value $B_{tot} = 3000$ min was selected as a moderate baseline at an intermediate budget level, where buffer allocations are still responsive to budget changes but already close to their saturation levels. Under this setting, average per-flight buffer levels range between 10 and 14 min.

Decisions are made prior to execution (pre-day-of-operations planning), and the model allocates proactive turnaround slack across flights that share aircraft rotation continuity through tail-number linkage.

The optimization results indicate substantial heterogeneity in turnaround buffer allocation across aircraft types and airline operating models.

Fig. 2 reports the mean optimized turnaround buffer by airline and aircraft type and shows that buffer allocation is primarily driven by daily aircraft utilization. Larger aircraft types tend to receive lower buffer allocations, which is consistent with their deployment on longer-haul services with fewer daily rotations, where delays are less likely to generate downstream effects across multiple flight legs. In contrast, smaller aircraft, typically operating multiple short-haul legs per day, receive lower buffers to mitigate the accumulation of delays over successive rotations. The results further suggest differences between full-service and low-cost carriers: low-cost airlines, which rely on tighter schedules and higher aircraft utilization, tend to receive lower buffers for comparable aircraft types, reflecting their greater exposure to downstream delays.

Table 2 provides context for these results by summarizing the number of flights and observation period associated with each airline-aircraft combination, highlighting differences in utilization intensity and operational scale. These variations in operational exposure influence the marginal value of proactive buffering under the assumed cost parameters.

Taken together, the evidence from Fig. 2 and Table 2 suggests that the proposed optimization framework allocates turnaround buffer selectively rather than uniformly. Buffer decisions emerge endogenously from the interaction between predicted turnaround uncertainty, aircraft

Table 1
Impact of scenario set size ($|\Omega|$) on solution stability and computational time.

Number of Scenarios $ \Omega $	Average Mean Buffer (min)	Runtime (sec)
5	9.907	86.8
10	10.029	134.9
20	10.031	260.2
40	9.97	438.2
60	9.911	545.9
80	9.546	699.4
100	9.574	918.4

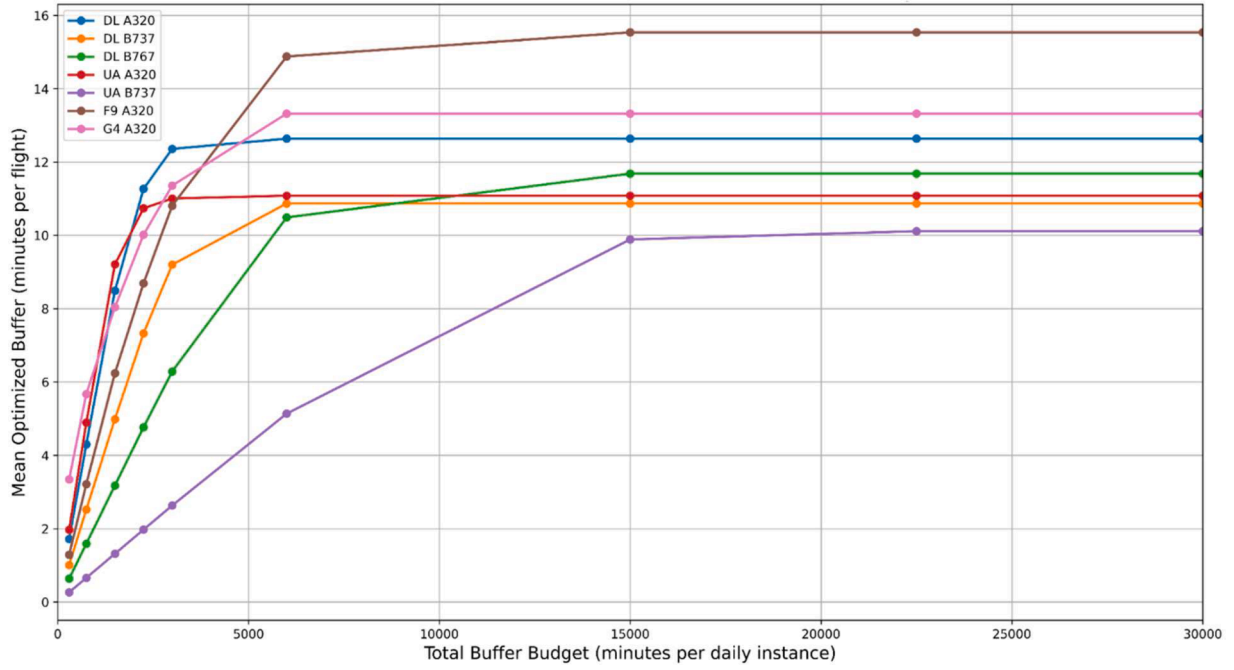


Fig. 1. Saturation behaviour of mean buffer allocation vs total budget.

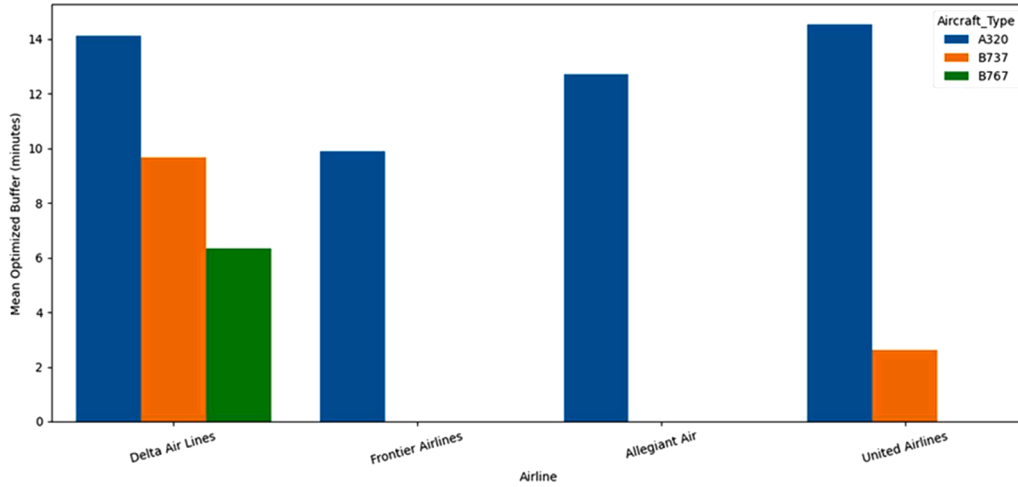


Fig. 2. Optimized turnaround buffer allocation by airline and aircraft type.

Table 2

Summary of optimized turnaround buffer allocation and sample characteristics.

Airline	Aircraft Type	Mean Buffer	N Flights	Days
Delta Airlines	A320	14.3	178	31
	B737	9.8	306	31
	B767	6.4	486	31
United Airlines	A320	14.9	157	31
	B737	2.6	1181	31
Frontier Airlines	A320	10.5	274	31
Allegiant Air	A320	12.6	186	31

utilization patterns, and the relative costs of planned slack versus realized delay, underscoring the importance of accounting for operational heterogeneity in proactive turnaround planning.

Solve times remain computationally lightweight. Per-instance solve times range from 0.040 s to 1.851 s, with a mean of 0.494 s and a 95th percentile of 1.616 s. The total runtime across all 217 instances is

approximately 337.53 s. All experiments were conducted using the CBC (COIN—OR Branch-and-Cut) solver through the PuLP modeling interface in Python, on a standard laptop (Intel i5–1135G7 CPU, 8GB RAM). These results demonstrate that the proposed predictive–prescriptive framework can be solved efficiently at operational scale for fleet-day instances involving up to approximately 1300 flights, without requiring specialized computational infrastructure.

We also compare the proposed stochastic optimization model against a zero buffer policy (no proactive slack), a uniform buffer policy in which the total daily budget B_{tot} is distributed equally across flights ($b_f = B_{tot}/F$), and a risk-proportional policy in which buffer time is allocated proportionally to the predicted propagated delay ($b_f \propto \hat{y}_f$). All policies are evaluated under identical simulation settings using $|\Omega| = 20$ scenarios with equal probabilities and the same cost parameters as the baseline optimization. Performance is assessed across the 217 airline, aircraft-type, and day instances in terms of expected total delay minutes, buffer cost, environmental proxy cost, and total objective value. Under the baseline budget $B_{tot} = 3000$ min, the comparative results are sum-

marized in Table 3.

The optimized model achieves a 37.1 % reduction in average departure delay minutes relative to the zero buffer policy, a 15.1 % reduction relative to uniform allocation, and a 4.7 % reduction relative to the risk-proportional policy. In terms of total objective value, the optimized solution improves upon zero buffer policy 25.6 %, upon uniform allocation by 13.9 % and upon the risk-proportional policy by 5.4 %. The environmental proxy cost is also lower than under both uniform and risk-proportional policies, reflecting improved delay absorption and reduced unnecessary ground occupancy. Also, the optimized allocation uses on average 2655 buffer minutes, below the full 3000-minute budget, indicating more efficient utilization of limited slack.

These results show that the prescriptive advantage of the proposed framework does not come simply from using delay predictions, but from optimally allocating buffers across all flights under a binding system-wide constraint. The stochastic optimization model aims to minimize buffer cost and environmental impact, and produces better out-of-sample results than uniform, proportional-to-risk, and observed-practice benchmark policies.

3.4. Sensitivity analysis

To assess the robustness of the optimization outcomes to economic assumptions, a sensitivity analysis was conducted by systematically varying the unit cost of delay and the unit cost of planned buffer time. This analysis examines how changes in the relative valuation of delay penalties and proactive slack influence the resulting turnaround buffer allocation, providing insight into the economic drivers of buffering decisions.

Fig. 3 presents the results of a sensitivity analysis with respect to the unit cost of delay and the unit cost of planned buffer time, where each cell reports the mean optimized buffer under a given cost combination. The results exhibit a clear and intuitive pattern: when delay costs are low relative to buffer costs, the model allocates little or no buffer, whereas buffer allocation increases rapidly as delay costs rise, reflecting the growing value of proactive slack in mitigating delay impacts. Higher buffer costs consistently suppress buffer allocation, even under high delay penalties, illustrating the fundamental trade-off between schedule efficiency and robustness.

The transition between low- and high-buffer regimes is nonlinear, indicating that modest changes in relative cost assumptions can lead to markedly different buffering strategies. Overall, the sensitivity analysis confirms that the proposed framework responds coherently to economic inputs and provides a flexible tool for exploring buffer policies under alternative cost scenarios.

4. Discussion

The results of this study highlight the importance of treating turnaround buffering as a proactive, flight-specific planning decision rather than a static or heuristic scheduling parameter. The optimization outcomes demonstrate that optimal buffer allocation varies systematically across aircraft types, utilization profiles, and airline operating models, reinforcing prior evidence that turnaround performance is closely tied to network structure and aircraft rotation patterns. By explicitly linking buffer decisions to predicted turnaround uncertainty, the proposed

framework provides a structured way to translate operational risk into actionable planning choices.

One key insight is that higher buffer allocations are not universally optimal, even under substantial turnaround uncertainty. Instead, the model selectively assigns buffer time to operations where the marginal benefit of proactive slack outweighs its opportunity cost in terms of reduced aircraft utilization. This finding aligns with empirical observations that delays propagate more strongly in high-frequency, short-haul operations, where aircraft perform multiple daily rotations and early disruptions can cascade across the schedule. Conversely, longer-haul and widebody operations, which typically involve fewer rotations and greater inherent slack, require less additional buffer to achieve comparable robustness.

The sensitivity analysis further underscores the economic nature of buffering decisions. The nonlinear transition between low- and high-buffer regimes indicates that relatively small changes in the valuation of delay penalties or buffer costs can lead to markedly different buffering strategies. From a managerial perspective, this highlights the value of using optimization-based tools to explore alternative cost scenarios rather than relying on fixed buffer rules derived from historical averages.

Beyond operational efficiency, the inclusion of an environmental proxy associated with extended total ground occupancy time illustrates how turnaround planning decisions can be aligned with broader sustainability objectives. While the environmental term plays a secondary role relative to delay and buffer costs, its presence encourages buffer allocations that balance robustness against prolonged ground operations and associated emissions. This is particularly relevant in the context of increasing regulatory and societal pressure on airlines to reduce ground emissions, especially at congested airports where APU usage is prevalent.

From an implementation standpoint, the proposed framework complements existing Airport Collaborative Decision Making (A-CDM) practices by shifting the focus upstream, from real-time coordination to proactive planning. Rather than reacting to realized delays, the model enables airlines to anticipate turnaround risk and allocate limited slack strategically across the schedule. This predictive-prescriptive integration is consistent with recent advances in data-driven airline operations and provides a practical bridge between machine learning-based prediction tools and operational decision-making.

Several limitations should be acknowledged. The analysis relies on historical operational data and assumes that future turnaround uncertainty follows similar patterns, which may not hold under structural changes such as major staffing disruptions or regulatory shifts. In addition, the current model treats buffer allocation as a single-stage planning problem and does not explicitly account for dynamic re-optimization as new information becomes available during the day of operations. Finally, while the environmental proxy captures emissions associated with total ground occupancy time, it does not account for broader network-level or passenger-related environmental impacts.

The proposed framework has been developed as a modular, post-scheduling decision support tool. As such, it does not explicitly model aircraft maintenance constraints, passenger connectivity considerations, or crew duty limitations within the optimization problem. Instead, these elements are assumed to be satisfied in the underlying schedule and can be enforced either upstream, by restricting the feasible solution space available to the buffer-allocation model, or downstream, by embedding

Table 3
Out-of-sample performance comparison of buffer allocation policies ($B_{t,0} = 3000$ min).

Policy	Avg dep. delay: $\delta_{f,0}$ (min)	Avg Buffer Cost: $c^{\text{buf}} b_f$	Avg Delay Cost: $c_t^{\text{delay}} \delta_{f,0}$	Avg Env Cost: $c^{\text{env}} g_{f,0}$	Avg Total Objective
Zero Buffer	5493.5	0	549,352	34,681	584,033
Uniform	4071.2	60	407,122	37,836	504,959
Risk-Proportional	3625.8	60	362,582	36,945	459,527
Optimized MILP	3455.3	53	345,532	35,915	434,556

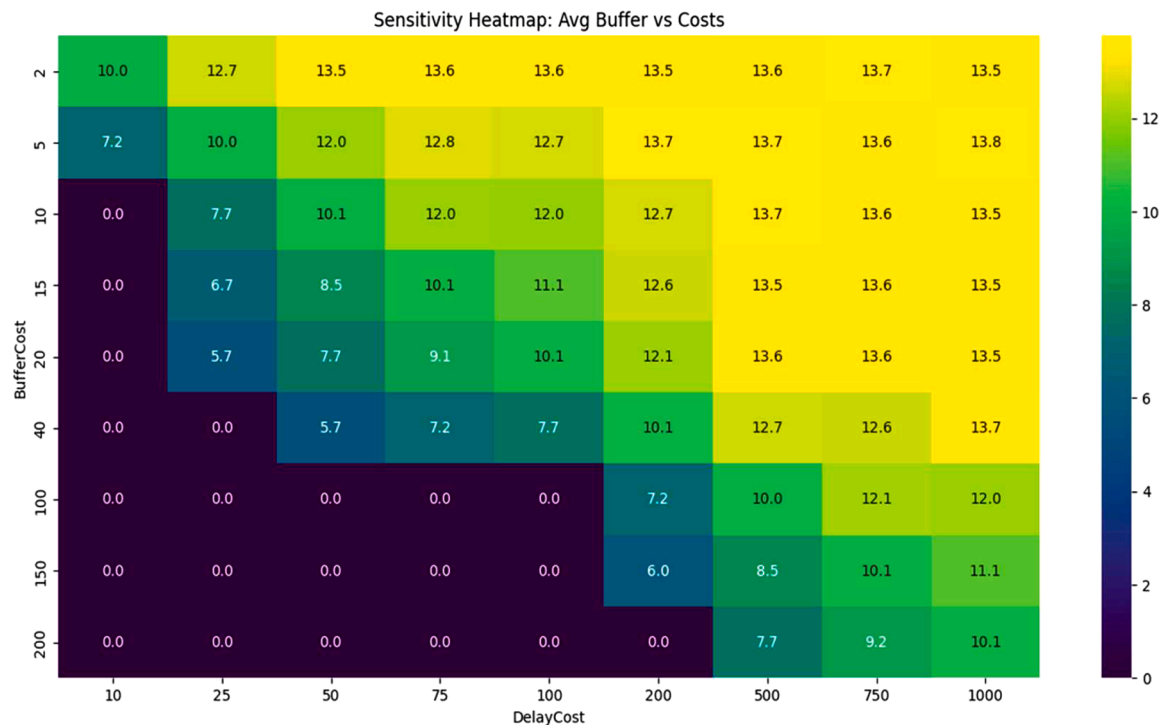


Fig. 3. Sensitivity heatmap of mean buffer versus delay and buffer costs.

the proposed approach within a broader integrated planning framework. Future research could extend the present model by incorporating these additional resource dimensions explicitly, or by coupling the buffer-allocation problem with integrated schedule design, fleet assignment, or crew pairing models. Such extensions would enable a more comprehensive evaluation of trade-offs across multiple operational layers, at the cost of increased model complexity and computational effort.

Despite these limitations, the results demonstrate that integrating machine-learning-derived uncertainty into optimization models can substantially enhance turnaround planning decisions. The framework provides airlines with a flexible and transparent tool for tailoring buffering strategies to their specific operational context, offering a meaningful step toward more robust, data-driven ground operations planning.

5. Conclusions

This paper proposed a learning-enhanced optimization framework for proactive aircraft turnaround buffer allocation that links flight-level delay predictions with prescriptive planning decisions. By embedding machine-learning-derived turnaround uncertainty into a stochastic optimization model, the framework allocates limited buffer time across flights to balance expected delay costs, buffer-related utilization costs, and system-level constraints. Results based on real operational data show that optimal buffering is highly heterogeneous and primarily driven by aircraft utilization patterns and exposure to delay accumulation, rather than by uniform rules or airline size. The sensitivity analysis further confirms that buffer allocations respond coherently to economic trade-offs between delay penalties and schedule efficiency, highlighting the value of proactive, data-driven buffering strategies over fixed or heuristic approaches.

Several avenues for future research emerge from this work. An important extension would be the incorporation of dynamic re-optimization, allowing buffer decisions to be updated as new operational information becomes available during the day of operations. Further research could also integrate turnaround buffering with other airline planning decisions, such as fleet assignment, crew scheduling, or

airport resource allocation, to capture broader network-level interactions. Expanding the framework to coordinate airport gate allocation and ground-resource limits, incorporate passenger connection constraints and crew legality constraints, or support aircraft reassignment and network rerouting would allow for integrated scheduling and recovery. From a modeling perspective, passenger connectivity could enhance the assessment of delay impacts, while more detailed environmental modeling could reflect emissions mitigation objectives. Although such extensions would substantially increase model dimensionality and computational complexity, they are conceptually compatible with the proposed framework and represent a natural next step toward embedding proactive turnaround buffering within large-scale integrated airline planning systems. Together, these extensions would further strengthen the role of predictive-prescriptive analytics in supporting robust and sustainable airline ground operations.

CRediT authorship contribution statement

Konstantia Kontodimou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Dionysios Peteinatos:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation. **Stavroula Bagioneta:** Writing – original draft, Methodology, Investigation, Conceptualization. **Konstantinos Kepaptsoglou:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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